EXPERIMENT 16: FEED FORWARD NEURAL NETWORK (FFNN) IMPLEMENTATION

AIM

To develop a Python program to implement a Feed Forward Neural Network (FFNN) for classification and predict the output based on the given input data.

OBJECTIVE

- To understand the concept of Feed Forward Neural Networks in Artificial Intelligence.
- To implement a neural network with an input layer, hidden layer, and output layer.
- To train the neural network using sample data.
- To perform classification using the trained network.
- To predict the output for new input data.

ALGORITHM

Step 1: Start.

Step 2: Prepare the input training data and corresponding target values.

Step 3: Create the Feed Forward Neural Network with:

- Input layer
- Hidden layer
- Output layer

Step 4: Initialize the weights and biases.

Step 5: Pass the input data forward through the network.

Step 6: Calculate the output using the activation function.

Step 7: Compare the predicted output with the actual output.

Step 8: Train the network by adjusting the weights and biases.

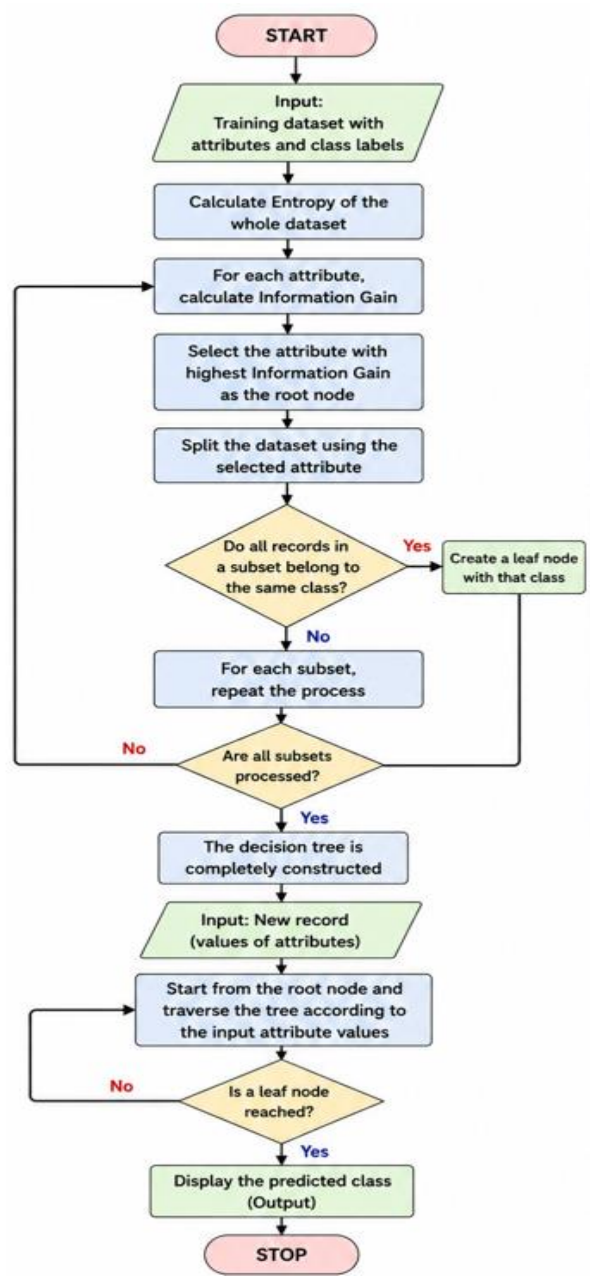
Step 9: Repeat the training process until the model learns the required pattern.

Step 10: Give new input data to the trained network.

Step 11: Predict and display the output.

Step 12: Stop.

FLOW CHART



SOURCE CODE

Feed Forward Neural Network

```
from sklearn.neural_network import MLPClassifier
```

```
# Training data
# [Input1, Input2]
X = [
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
]

# Target output (AND gate)
y = [0, 0, 0, 1]

# Create Feed Forward Neural Network
model = MLPClassifier(
    hidden_layer_sizes=(4,),
    activation='relu',
    solver='lbfgs',
    max_iter=1000,
    random_state=1
)

# Train the neural network
model.fit(X, y)

# Test input
input1 = int(input("Enter first input (0 or 1): "))
input2 = int(input("Enter second input (0 or 1): "))
```

```
# Predict output
prediction = model.predict([[input1, input2]])


print("Predicted Output:", prediction[0])
```

OUTPUT

```
Enter first input (0 or 1): 0
Enter second input (0 or 1): 0
Predicted Output: 0
```

RESULT

The Python program successfully implements a Feed Forward Neural Network and predicts the output for the given input data after training the network on the provided dataset.